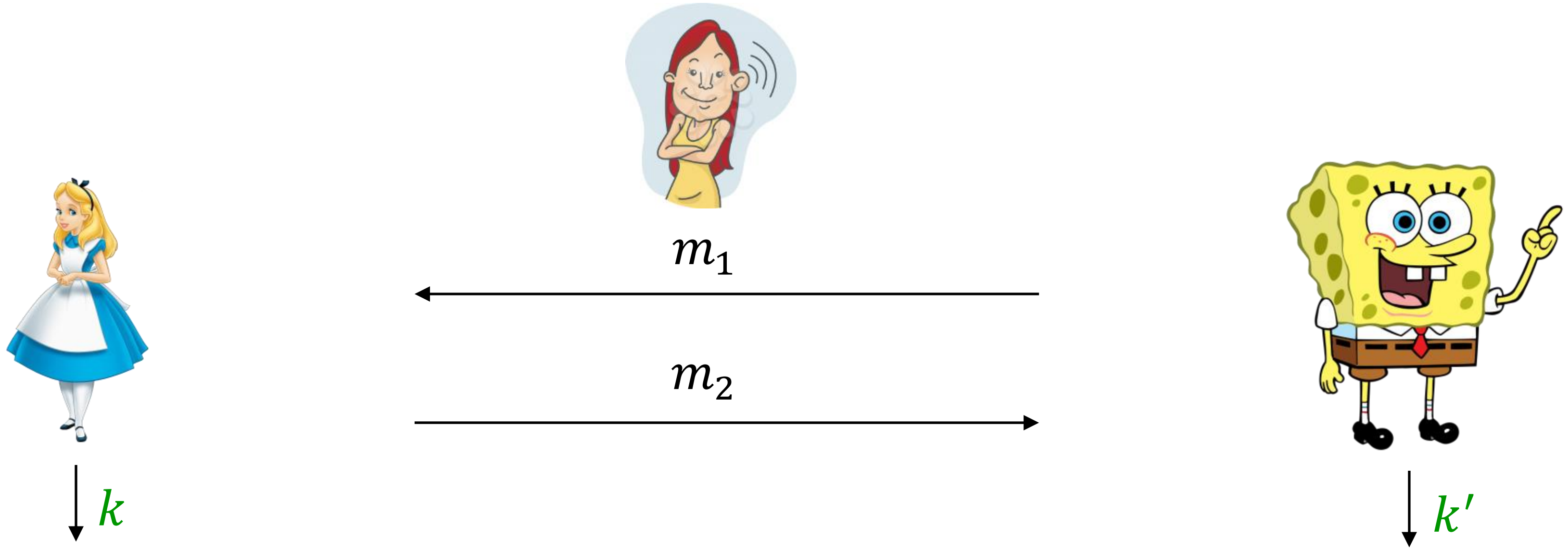
COS 433: Cryptography

Announcements:

- Problem Set 3 due **10/29**
- Problem Set 4 to be released **10/29. Due** Wed 11/12

Discrete Log and Diffie-Hellman

Key Agreement



- Correctness: if Alice and Bob behave honestly, then $k = k'$
- Weak security: hard for  to find k given (m_1, m_2).
- Strong security: $(m_1, m_2, k) \approx_c (m_1, m_2, u)$ (uniformly random key u)

The Diffie-Hellman Proposal

Tool: $\mathbb{Z}_p$-discrete logarithm problem

- Hard problem defined over $\mathbb{Z}_p$ instead of $\mathbb{Z}_N$
- Number theory background:
  - $\mathbb{Z}_p^\times = \{1, 2, \dots, p-1\}$ is a multiplicative group
  - Theorem: $\mathbb{Z}_p^\times$ is cyclic
    - There exists a generator $g \in \mathbb{Z}_p^\times$ such that $\mathbb{Z}_p^\times = \{1, g, g^2, \dots, g^{p-2}\}$

Discrete log problem: given $(g, g^x \bmod p)$, find x

Discrete log assumption over $\mathbb{Z}_p^\times$

Assumption: parametrized by a prime p and generator g

For all poly-time A,

$$\Pr_{x \leftarrow \mathbb{Z}_{p-1}}[A(p, g, g^x) = x] = \mathrm{negl}(\lambda)$$

- Where do (p, g) come from?
  - p can be sampled uniformly at random from $\{0,1\}^\lambda$ (check primality)
  - Sample g uniformly? Unclear how to tell if g is a generator

Discrete log assumption over $\mathbb{Z}_p^\times$

Assumption: parametrized by a prime p and generator g

For all poly-time A,

$$\Pr_{x \leftarrow \mathbb{Z}_{p-1}} [A(p, g, g^x) = x] = \mathrm{negl}(\lambda)$$

- Where do (p, g) come from?
  - Sample $p - 1$ to be a uniformly random factored integer (tricky)
    - Check if p is prime. Else try again.
  - Sample g uniformly and check that $g^{\frac{p-1}{q}} \neq 1$ for all prime $q | p - 1$

Discrete log assumption over $\mathbb{Z}_p^\times$

Assumption (random p, g): for all poly-time A,

$$\Pr_{\substack{p \leftarrow \{0,1\}^\lambda,\ p\ \text{prime} \\ g \leftarrow \mathbb{Z}_p^\times,\ g\ \text{generator} \\ x \leftarrow \mathbb{Z}_{p-1}}} [A(p, g, g^x) = x] = \mathrm{negl}(\lambda)$$

- Implies the existence of one-way functions
  - $f(p, g, x) = (p, g, g^x)$ (up to domain sampling tricks)
- Not known to imply PKE
- We will make stronger assumptions.

Diffie-Hellman Key Agreement

p, g known (or sampled by Bob)

$$m_1 = g^{x}$$

$$\longleftarrow$$

$$m_2 = g^{y}$$

$$\longrightarrow$$

$y \leftarrow \mathbb{Z}_{p-1}$

$\downarrow$

$k = m_1^{y}$

$x \leftarrow \mathbb{Z}_{p-1}$

$\downarrow$

$k' = m_2^{x}$

- Correctness:

$$k = m_1^{y} \equiv g^{xy} \equiv m_2^{x} = k'$$

Diffie-Hellman Key Agreement

p, g known (or sampled by Bob)

$$\xleftarrow{\quad m_1 = g^{x} \quad}$$

$$\xrightarrow{\quad m_2 = g^{y} \quad}$$

$y \leftarrow \mathbb{Z}_{p-1}$

$\downarrow$

$k = m_1^{y}$

$x \leftarrow \mathbb{Z}_{p-1}$

$\downarrow$

$k' = m_2^{x}$

- Security?
  - If discrete logarithm is easy, then this is broken.

Computational Diffie-Hellman Assumption over $\mathbb{Z}_p^\times$

Assumption parametrized by a prime p and generator g (usually sampled at random): For all poly-time A,

$$\Pr_{x,y \leftarrow \mathbb{Z}_{p-1}}[A(p, g, g^x, g^y) = g^{xy}] = \mathrm{negl}(\lambda)$$

- Implies the weak security of Diffie-Hellman key agreement
- Open: show this is equivalent to the hardness of discrete log

What about strong security?

Decisional Diffie-Hellman Assumption over $\mathbb{Z}_p^\times$

Assumption parametrized by a prime p and generator g (usually sampled at random):

$$(p, g, g^x, g^y, g^{xy}) \approx_c (p, g, g^x, g^y, g^z)$$

- Implies the strong security of Diffie-Hellman key agreement
- False!
- g^{xy} is a square mod p with probability ¾; g^z is with probability ½

Decisional Diffie-Hellman Assumption over $\mathbb{Z}_p^\times$

Assumption parametrized by a prime p and generator g (usually sampled at random):

$$(p, g, g^x, g^y, g^{xy}) \approx_c (p, g, g^x, g^y, g^z)$$

Now what?

Decisional Diffie-Hellman Assumption over G

Assumption: parametrized by a cyclic group G and generator g (usually sampled from a distribution)

$$(\langle G \rangle, g, g^{x}, g^{y}, g^{xy}) \approx_c (\langle G \rangle, g, g^{x}, g^{y}, g^{z})$$

Solution: we don't have to work over $\mathbb{Z}_p$!

Decisional Diffie-Hellman Assumption over G

Assumption: parametrized by a cyclic group G and generator g (usually sampled from a distribution)

$$(\langle G \rangle, g, g^{x}, g^{y}, g^{xy}) \approx_c (\langle G \rangle, g, g^{x}, g^{y}, g^{z})$$

What we need: a "family of effective groups"

- Group $G^{(\lambda)}$ for each λ (or a distribution over groups)
- $\mathrm{poly}(\lambda)$ time group operations
- Easy to sample a group along with a generator g for $G^{(\lambda)}$

Decisional Diffie-Hellman Assumption over G

Assumption: parametrized by a cyclic group G and generator g (usually sampled from a distribution)

$$(\langle G\rangle, g, g^{x}, g^{y}, g^{xy}) \approx_c (\langle G\rangle, g, g^{x}, g^{y}, g^{z})$$

In what kinds of groups is DDH plausible?

- Need discrete log to be hard
- Need G not to have small subgroups.

Decisional Diffie-Hellman Assumption over G

Assumption: parametrized by a cyclic group G and generator g (usually sampled from a distribution)

$$(\langle G\rangle, g, g^{x}, g^{y}, g^{xy}) \approx_c (\langle G\rangle, g, g^{x}, g^{y}, g^{z})$$

In what kinds of groups is DDH plausible?

- Simple idea: focus on G such that $|G| = q$ is a prime!

Decisional Diffie-Hellman Assumption over G

Assumption: parametrized by a cyclic group G and generator g (usually sampled from a distribution)

$$(\langle G\rangle, g, g^{x}, g^{y}, g^{xy}) \approx_c (\langle G\rangle, g, g^{x}, g^{y}, g^{z})$$

In what kinds of groups is DDH plausible?

- QR_p where $\frac{p-1}{2}$ is a prime. ("Safe Primes")
- QR_N where $N = pq$ and p, q are safe primes (not prime order).

Decisional Diffie-Hellman Assumption over G

Assumption: parametrized by a cyclic group G and generator g (usually sampled from a distribution)

$$(\langle G\rangle, g, g^{x}, g^{y}, g^{xy}) \approx_c (\langle G\rangle, g, g^{x}, g^{y}, g^{z})$$

In what kinds of groups is DDH plausible?

- QR_p where $\frac{p-1}{2}$ is a prime. ("Safe Primes")
- (Prime order) "elliptic curve groups"

Diffie-Hellman Key Agreement

$G, L = |G|, g$ known or sampled by Bob

$$m_1 = g^{x} \quad \longleftarrow$$

$$m_2 = g^{y} \quad \longrightarrow$$

Alice side: $y \leftarrow \mathbb{Z}_L$, then $k = m_1^{y}$

Bob side: $x \leftarrow \mathbb{Z}_L$, then $k' = m_2^{x}$

Theorem:

- If CDH is true for $(\langle G \rangle, g)$, then protocol is weakly secure
- If DDH is true for $(\langle G \rangle, g)$, then protocol is strongly secure

Diffie-Hellman Key Agreement

$G, L = |G|, g$ known or sampled by Bob

$m_1 = g^x$ ←

$m_2 = g^y$ →

$y \leftarrow \mathbb{Z}_L$ ↓ $k = m_1^y$

$x \leftarrow \mathbb{Z}_L$ ↓ $k' = m_2^x$

Note: you don't even need to know L to run this (details omitted)

ElGamal Encryption

$G, L = |G|, g$ known or sampled by Bob (part of pk)

h

$\mathrm{ct} = g^y, h^y \cdot m$

$m' = \mathrm{ct}_2 \cdot \mathrm{ct}_1^{-x}$

$m \in \{0,1\}^n \subset G$
$y \leftarrow \mathbb{Z}_L$

$x \leftarrow \mathbb{Z}_L$
$\begin{pmatrix} \mathrm{pk} = h = g^x, \\ \mathrm{sk} = x \end{pmatrix}$

PKE based on the DDH assumption; "encryption version" of Diffie-Hellman protocol

- Correctness: $h^y = (g^y)^x$
- Lemma: if the DDH assumption holds, then ElGamal encryption is secure.
- Note: unlike our other schemes, this does not use a trapdoor permutation

Algorithms for Discrete Log (general groups)

1. Brute force

- Running time: $|G| \cdot \mathbf{poly}(\lambda) = 2^{\lambda} \cdot \mathbf{poly}(\lambda)$ for $|G| \approx 2^{\lambda}$

2. Baby Step—Giant Step

- Let $|G| \leq B^2$ (think $B = 2^{\lambda/2}$)
- Given $(g, h = g^x)$, solve for $0 \leq y, z \leq B$ such that

$$h = g^{y \cdot B + z}$$

- Enumerate all $g^{y \cdot B}, h \cdot g^{-z}$. Look for a collision!
- Running time $= B \cdot \mathbf{poly}(\lambda)$. Quadratic speedup $(2^{\lambda/2})$ over brute force.

3. That's all we know!

(Certain) Elliptic curve groups: can set $\lambda = 256$ (pretty small!)

Algorithms for Discrete Log ($\mathbb{Z}_p^\times$ and subgroups)

1. Brute force
    - Running time: $|G| \cdot \mathbf{poly}(\lambda) = 2^\lambda \cdot \mathbf{poly}(\lambda)$ for $|G| \approx 2^\lambda$
2. Baby Step—Giant Step
    - Running time $= B \cdot \mathbf{poly}(\lambda)$. Quadratic speedup $(2^{\lambda/2})$ over brute force.
3. Index Calculus Algorithms
    - Use random-self reduction: $\mathbf{dlog}(g^x) = \mathbf{dlog}(g^{x+r}) - r$
    - Hope that g^{x+r} is "special," develop a tailored algorithm for "special" h
    - Since over $\mathbb{Z}_p^\times$, g^{x+r} is a random integer, one can show it is often special
        - Quadratic Sieve runs in time $2^{O(\sqrt{\lambda \cdot \log \lambda})}$
        - Number Field Sieve runs heuristically in time $2^{O\left(\sqrt[3]{\lambda \cdot \log^2(\lambda)}\right)}$

Algorithms for CDH (any group)

1. Solve discrete log over G

2. That's basically all we know!

Algorithms for DDH (any group)

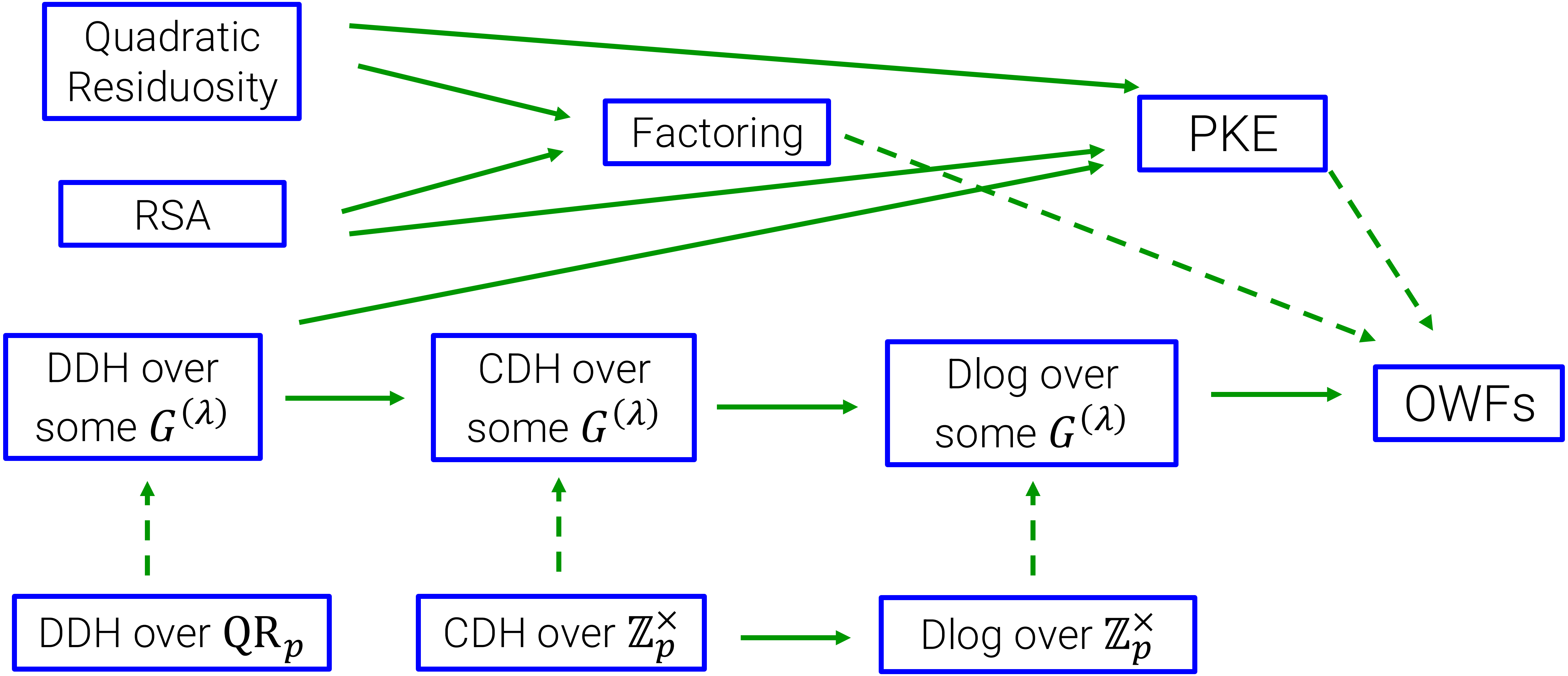
1. Solve discrete log over G

2. Solve discrete log over subgroups of G
   - Breaks certain G such as $\mathbb{Z}_p^\times$

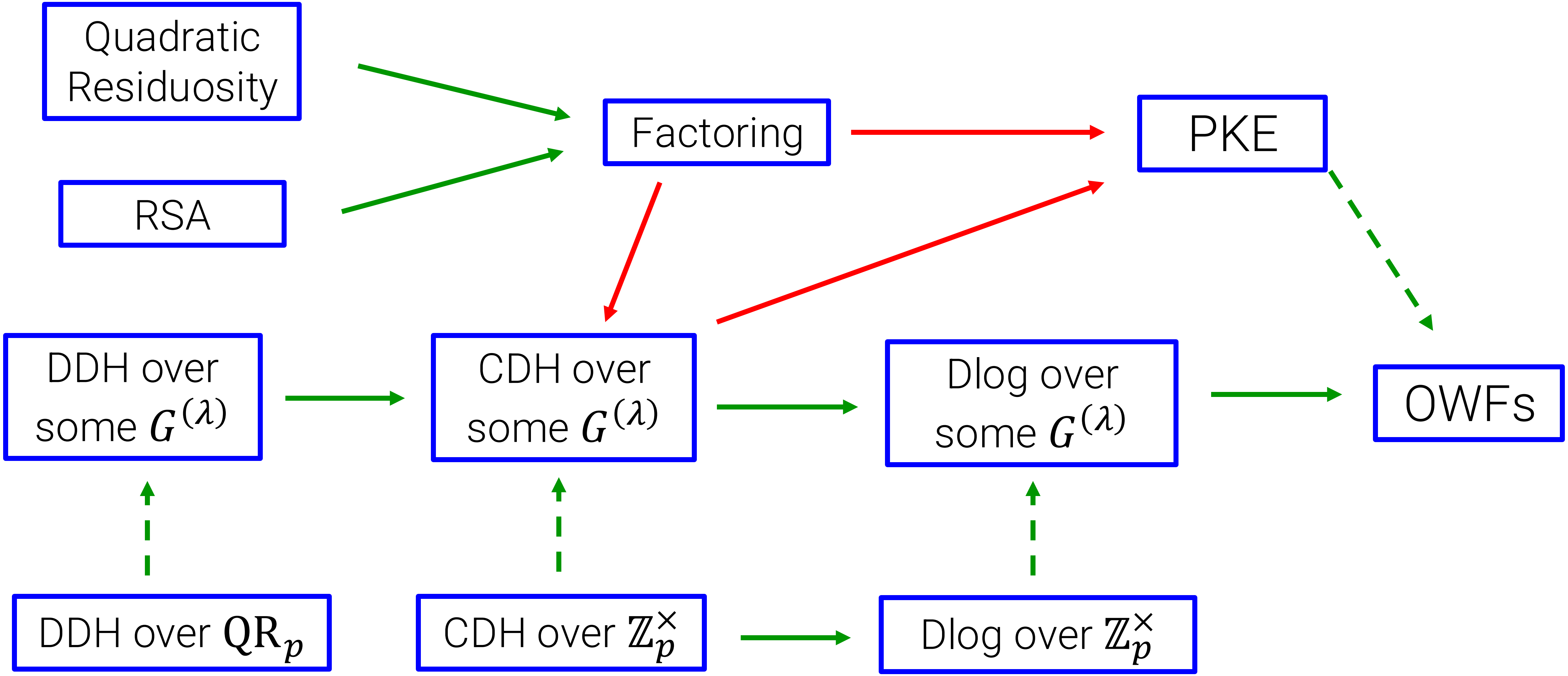
3. It's complicated…
   - DDH broken over certain weird/interesting groups ("self-pairings")

Nevertheless, groups such as $\mathbf{QR}_p$ have no known algorithms beyond (1)

Number-Theoretic Hardness Assumptions



Number-Theoretic Hardness Assumptions



End of PKE discussion (for now)

Next: digital signatures (authentication)